

PERTEMUAN 12

AKAR-AKAR PERSAMAAN KARAKTERISTIK DENGAN METODE REGULAR FALSI

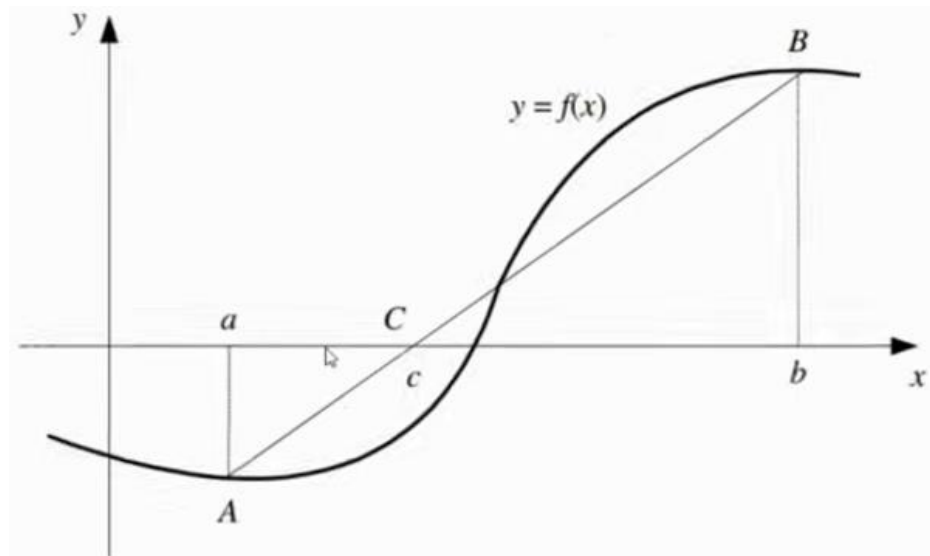
A. Capaian Pembelajaran

Setelah mengikuti pertemuan ini, mahasiswa diharapkan mampu memahami dan menyelesaikan permasalahan pada soal-soal matematika khususnya persamaan non-linier dengan menggunakan metode Regular Falsi.

B. Uraian Materi

1. Metode Regular Falsi

Metode Regular Falsi atau juga yang dikenal dengan metode posisi palsu adalah metode yang pengoperasiannya memiliki kemiripan dengan metode biseksi (metode bagi dua) dimana metode ini juga melakukan langkah-langkah peng-iterasian secara terus menerus sembari melakukan pembaharuan nilai pada rentang-rentang nya guna mendapatkan nilai dari akar persamaan yang dicari.



Gambar Grafik Regular Falsi

Langkah-langkah dalam penggunaan metode regular falsi:

1. Menentukan selang $[a, b]$

2. Menghitung $c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$

3. Apabila

- $f(a).f(c) < 0$ maka selang baru $[a, c] \rightarrow b = c$
- $f(c).f(b) < 0$ maka selang baru $[c, b] \rightarrow a = c$
- Kunci terpenting adalah nilai di atas nol pasti positif di bawah nol pasti negatif, sehingga pencarian nilai selang baru dapat menjadi lebih mudah dieksekusi.

4. Iterasi berhenti apabila $f(c) = 0$

Contoh Soal

Tentukan salah satu akar persamaan berikut

$f(x) = x^2 - 6x + 5 = 0$ dengan selang $[3,6]$ dengan tingkat ketelitian 3 *desimal*

Jawab

$$a = 3, b = 6$$

iterasi 1

$$f(a) = a^2 - 6a + 5$$

$$\begin{aligned} f(3) &= 3^2 - 6(3) + 5 \\ &= 9 - 18 + 5 \\ &= -4 \end{aligned}$$

$$f(b) = b^2 - 6b + 5$$

$$\begin{aligned} f(6) &= 6^2 - 6(6) + 5 \\ &= 36 - 36 + 5 \\ &= 5 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{5.3 - (-4).6}{5 - (-4)}$$

$$c = 4.333$$

$$f(c) = c^2 - 6c + 5$$

$$\begin{aligned} f(4.333) &= 4.333^2 - 6(4.333) + 5 \\ &= -2.222 \end{aligned}$$

$$f(a).f(c) = -4 \times (-2.222) = 8.889$$

$$f(c).f(b) = -2.222 \times 5 = -11.111$$

$$f(c).f(b) = -11.111 < 0, a = c = 4.333$$

$$\text{Galat relatif} = 4.333$$

iterasi 2

$$f(a) = a^2 - 6a + 5$$

$$\begin{aligned} f(4.333) &= 4.333^2 - 6(4.333) + 5 \\ &= -2.222 \end{aligned}$$

$$f(b) = b^2 - 6b + 5$$

$$\begin{aligned} f(6) &= 6^2 - 6(6) + 5 \\ &= 36 - 36 + 5 \\ &= 5 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{5.(4.333) - (-2.222).6}{5 - (-2.222)}$$

$$c = 4.846$$

$$f(c) = c^2 - 6c + 5$$

$$\begin{aligned} f(4.846) &= 4.846^2 - 6(4.846) + 5 \\ &= -0.592 \end{aligned}$$

$$f(a).f(c) = (-2.222) \times (-0.592) = 1.315$$

$$f(c).f(b) = (-0.592) \times 5 = -2.959$$

$$f(c).f(b) = -2.959 < 0, a = c$$

$$\text{Galat relatif} = 0.105$$

iterasi 3

$$f(a) = a^2 - 6a + 5$$

$$\begin{aligned} f(4.846) &= 4.846^2 - 6(4.846) + 5 \\ &= -0.592 \end{aligned}$$

$$f(b) = b^2 - 6b + 5$$

$$\begin{aligned} f(6) &= 6^2 - 6(6) + 5 \\ &= 36 - 36 + 5 \\ &= 5 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{5.(4.846) - (-0.592).6}{5 - (-0.592)}$$

$$c = 4.968$$

$$f(c) = c^2 - 6c + 5$$

$$\begin{aligned} f(4.968) &= 4.968^2 - 6(4.968) + 5 \\ &= -0.126 \end{aligned}$$

$$f(a).f(c) = (-0.592) \times (-0.126) = 0.075$$

$$f(c).f(b) = (-0.126) \times 5 = -0.630$$

$$f(c).f(b) = -0.630 < 0, a = c$$

$$\text{Galat relative} = 0.024$$

iterasi 4

$$f(a) = a^2 - 6a + 5$$

$$\begin{aligned} f(4.968) &= 4.968^2 - 6(4.968) + 5 \\ &= -0.126 \end{aligned}$$

$$f(b) = b^2 - 6b + 5$$

$$\begin{aligned} f(6) &= 6^2 - 6(6) + 5 \\ &= 36 - 36 + 5 \\ &= 5 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{5.(4.968) - (-0.126).6}{5 - (-0.126)}$$

$$c = 4.994$$

$$f(c) = c^2 - 6c + 5$$

$$\begin{aligned} f(4.994) &= 4.994^2 - 6(4.994) + 5 \\ &= -0.026 \end{aligned}$$

$$f(a).f(c) = (-0.126) \times (-0.026) = 0.003$$

$$f(c).f(b) = (-0.026) \times 5 = -0.128$$

$$f(c).f(b) = -0.128 < 0, a = c$$

$$\text{Galat relative} = 0.005$$

iterasi 5

$$f(a) = a^2 - 6a + 5$$

$$\begin{aligned} f(4.994) &= 4.994^2 - 6(4.994) + 5 \\ &= -0.026 \end{aligned}$$

$$f(b) = b^2 - 6b + 5$$

$$\begin{aligned} f(6) &= 6^2 - 6(6) + 5 \\ &= 36 - 36 + 5 \\ &= 5 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{5.(4.994) - (-0.026).6}{5 - (-0.026)}$$

$$c = 4.999$$

$$f(c) = c^2 - 6c + 5$$

$$\begin{aligned} f(4.999) &= 4.999^2 - 6(4.999) + 5 \\ &= -0.004 \end{aligned}$$

$$f(a).f(c) = (-0.026) \times (-0.004) = 0.0001$$

$$f(c).f(b) = (-0.004) \times 5 = -0.025$$

$$f(c).f(b) = -0.0255 < 0, a = c$$

$$\text{Galat relative} = 0.001$$

iterasi 6

$$f(a) = a^2 - 6a + 5$$

$$\begin{aligned} f(4.999) &= 4.999^2 - 6(4.999) + 5 \\ &= -0.004 \end{aligned}$$

$$f(b) = b^2 - 6b + 5$$

$$\begin{aligned} f(6) &= 6^2 - 6(6) + 5 \\ &= 36 - 36 + 5 \\ &= 5 \end{aligned}$$

$$c = \frac{f(b) \cdot a - f(a) \cdot b}{f(b) - f(a)}$$

$$c = \frac{5 \cdot (4.999) - (-0.004) \cdot 6}{5 - (-0.004)}$$

$$c = 5$$

$$f(c) = c^2 - 6c + 5$$

$$f(5) = 5^2 - 6(5) + 5$$

$$= 0$$

Kesimpulan, solusi $x = c = 5$ dengan $f(c) = 0$

Contoh Soal

Tentukan salah satu akar persamaan berikut

$$f(x) = 2x^2 - 3x - 5 = 0$$

dengan selang $[0,4]$ dengan tingkat ketelitian 0.002!

Jawab

$$a = 0, b = 4$$

iterasi 1

$$f(a) = 2a^2 - 3a - 5$$

$$f(0) = 2(0)^2 - 3(0) - 5$$

$$= 0 - 0 - 5$$

$$= -5$$

$$f(b) = 2b^2 - 3b - 5$$

$$f(4) = 2(4)^2 - 3(4) - 5$$

$$= 32 - 12 - 5$$

$$= 15$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{(15(0)) - (-5(4))}{15 - (-5)}$$

$$c = 1$$

$$f(c) = 2c^2 - 3c - 5$$

$$\begin{aligned} f(1) &= 2(1)^2 - 3(1) - 5 \\ &= -6 \end{aligned}$$

$$f(a).f(c) = -5 \times (-6) = 30$$

$$f(c).f(b) = -6 \times 15 = -90$$

$$f(c).f(b) = -90 < 0, a = c$$

iterasi 2

$$f(a) = 2a^2 - 3a - 5$$

$$\begin{aligned} f(1) &= 2(1)^2 - 3(1) - 5 \\ &= -6 \end{aligned}$$

$$f(b) = 2b^2 - 3b - 5$$

$$\begin{aligned} f(4) &= 2(4)^2 - 3(4) - 5 \\ &= 15 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{(15(1)) - (-5(4))}{15 - (-6)}$$

$$c = 1.85714$$

$$f(c) = 2c^2 - 3c - 5$$

$$\begin{aligned} f(1.85714) &= 2(1.85714)^2 - 3(1.85714) - 5 \\ &= -3.67346 \end{aligned}$$

$$f(a).f(c) = -6 \times (-3.57346) = 22.04081$$

$$f(c).f(b) = -3.67346 \times 15 = -55.102$$

$$f(c).f(b) = -55.102 < 0, a = c$$

$$\text{Galat relative} = 0.461538$$

iterasi 3

$$f(a) = 2a^2 - 3a - 5$$

$$\begin{aligned} f(1.85714) &= 2(1.85714)^2 - 3(1.85714) - 5 \\ &= -3.67346 \end{aligned}$$

$$f(b) = 2b^2 - 3b - 5$$

$$\begin{aligned} f(4) &= 2(4)^2 - 3(4) - 5 \\ &= 15 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{(15(1.85714)) - (-3.67346(4))}{15 - (-3.67346)}$$

$$c = 2.27868$$

$$f(c) = 2c^2 - 3c - 5$$

$$\begin{aligned} f(2.27868) &= 2(2.27868)^2 - 3(2.27868) - 5 \\ &= -1.45122 \end{aligned}$$

$$f(a).f(c) = -3.67346 \times (-1.45122) = 5.33102$$

$$f(c).f(b) = -1.45122 \times 15 = -21.76834$$

$$f(c).f(b) = -21.76834 < 0, a = c$$

$$\text{Galat relative} = 0.184994$$

iterasi 4

$$f(a) = 2a^2 - 3a - 5$$

$$\begin{aligned} f(2.27868) &= 2(2.27868)^2 - 3(2.27868) - 5 \\ &= -1.45122 \end{aligned}$$

$$f(b) = 2b^2 - 3b - 5$$

$$\begin{aligned} f(4) &= 2(4)^2 - 3(4) - 5 \\ &= 15 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{(15(2.27868)) - (-1.45122(4))}{15 - (-1.45122)}$$

$$c = 2.43053$$

$$f(c) = 2c^2 - 3c - 5$$

$$\begin{aligned} f(2.43053) &= 2(2.43053)^2 - 3(2.43053) - 5 \\ &= -0.47662 \end{aligned}$$

$$f(a).f(c) = -1.45122 \times (-0.47662) = 0.69169$$

$$f(c).f(b) = -0.47662 \times 15 = -7.14939$$

$$f(c).f(b) = -7.14939 < 0, a = c$$

$$\text{Galat relative} = 0.06247$$

iterasi 5

$$f(a) = 2a^2 - 3a - 5$$

$$\begin{aligned} f(2.43053) &= 2(2.43053)^2 - 3(2.43053) - 5 \\ &= -0.47662 \end{aligned}$$

$$f(b) = 2b^2 - 3b - 5$$

$$\begin{aligned} f(4) &= 2(4)^2 - 3(4) - 5 \\ &= 15 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{(15(2.43053)) - (-0.47662(4))}{15 - (-0.47662)}$$

$$c = 2.47886$$

$$f(c) = 2c^2 - 3c - 5$$

$$\begin{aligned} f(2.47886) &= 2(2.47886)^2 - 3(2.47886) - 5 \\ &= -0.14704 \end{aligned}$$

$$f(a).f(c) = -0.47662 \times (-0.14704) = 0.070085$$

$$f(c).f(b) = -0.14704 \times 15 = -2.20568$$

$$f(c).f(b) = -2.20568 < 0, a = c$$

$$\text{Galat relative} = 0.01949$$

iterasi 6

$$f(a) = 2a^2 - 3a - 5$$

$$\begin{aligned} f(2.47886) &= 2(2.47886)^2 - 3(2.47886) - 5 \\ &= -0.14704 \end{aligned}$$

$$f(b) = 2b^2 - 3b - 5$$

$$\begin{aligned} f(4) &= 2(4)^2 - 3(4) - 5 \\ &= 15 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{(15(2.47886)) - (-0.14704(4))}{15 - (-0.14704)}$$

$$c = 2.49363$$

$$f(c) = 2c^2 - 3c - 5$$

$$\begin{aligned} f(2.49363) &= 2(2.49363)^2 - 3(2.49363) - 5 \\ &= -0.04448 \end{aligned}$$

$$f(a).f(c) = -0.14704 \times (-0.04448) = 0.00654$$

$$f(c).f(b) = -0.04448 \times 15 = -0.66733$$

$$f(c).f(b) = -0.66733 < 0, a = c$$

$$\text{Galat relative} = 0.005921$$

iterasi 7

$$f(a) = 2a^2 - 3a - 5$$

$$\begin{aligned} f(2.49363) &= 2(2.49363)^2 - 3(2.49363) - 5 \\ &= -0.04448 \end{aligned}$$

$$f(b) = 2b^2 - 3b - 5$$

$$\begin{aligned} f(4) &= 2(4)^2 - 3(4) - 5 \\ &= 15 \end{aligned}$$

$$c = \frac{f(b).a - f(a).b}{f(b) - f(a)}$$

$$c = \frac{(15(2.49363)) - (-0.04448(4))}{15 - (-0.04448)}$$

$$c = 2.49808$$

$$f(c) = 2c^2 - 3c - 5$$

$$\begin{aligned} f(2.49808) &= 2(2.49808)^2 - 3(2.49808) - 5 \\ &= -0.01338 \end{aligned}$$

$$f(a).f(c) = -0.04448 \times (-0.01338) = 0.00059$$

$$f(c).f(b) = -0.01338 \times 15 = -0.200711$$

Galat relative = 0.00178 , iterasi dihentikan karena sudah mencapai ketentuan tingkat ketelitian

C. Latihan Soal

4. Dengan menggunakan metode regular falsi, temukan akar persamaan di bawah ini dimana selang $[1, 3]$ dengan tingkat ketelitian 0.001!

$$\frac{2}{2 + 2x}$$

5. Buktikan bahwa metode regular falsi mendapatkan hasil akar lebih cepat dengan membandingkan bersama metode biseksi pada persamaan fungsi dibawah ini dengan selang $[0, 3]$ dan tingkat ketelitian 0.005!

$$2x^3 + 2x^2 + 2x$$

D. Daftar Pustaka

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